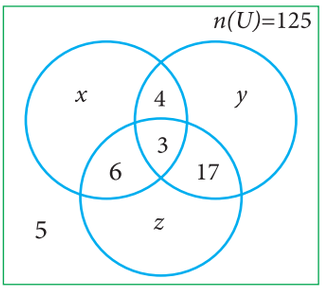
(i) Number of families buy only one newspaper

(ii) Number of families buy atleast two newspapers

(iii) Total number of families in the colony.

10. A survey of 1000 farmers found that 600 grew paddy, 350 grew ragi, 280 grew corn, 120 grew paddy and ragi, 100 grew ragi and corn, 80 grew paddy and corn. If each farmer grew atleast any one of the above three, then find the number of farmers who grew all the three.



11. In the adjacent diagram, if $n(\mathrm{U}) = 125$, y is two times of x and z is 10 more than x, then find the value of x, y and z.

12. Each student in a class of 35 plays atleast one game among chess, carrom and table tennis. 22 play chess, 21 play carrom, 15 play table tennis, 10 play chess and table tennis, 8 play carrom and table tennis and 6 play all the three games. Find the number of students who play (i) chess and carrom but not table tennis (ii) only chess (iii) only carrom (**Hint:** Use Venn diagram)

13. In a class of 50 students, each one come to school by bus or by bicycle or on foot. 25 by bus, 20 by bicycle, 30 on foot and 10 students by all the three. Now how many students come to school exactly by two modes of transport?

Exercise 1.7

TQPG8

Multiple Choice Questions

1. Which of the following is correct?

(1) $\{7\} \in \{1,2,3,4,5,6,7,8,9,10\}$ (2) $7 \in \{1,2,3,4,5,6,7,8,9,10\}$

(3) $7 \notin \{1,2,3,4,5,6,7,8,9,10\}$ (4) $\{7\} \not\subseteq \{1,2,3,4,5,6,7,8,9,10\}$

2. The set $P = \{x \mid x \in \mathbb{Z}, -1 < x < 1\}$ is a

(1) Singleton set (2) Power set (3) Null set (4) Subset

3. If $\mathrm{U} = \{x \mid x \in \mathbb{N}, x < 10\}$ and $A = \{x \mid x \in \mathbb{N}, 2 \le x < 6\}$ then $(A')'$ is

(1) {1, 6, 7, 8, 9} (2) {1, 2, 3, 4} (3) {2, 3, 4, 5} (4) { }

4. If $B \subseteq A$ then $n(A \cap B)$ is

(1) $n(A-B)$ (2) $n(B)$ (3) $n(B - A)$ (4) $n(A)$

5. If $A = \{x, y, z\}$ then the number of non- empty subsets of A is

(1) 8 (2) 5 (3) 6 (4) 7

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